Personalised Practice: Resultant Forces

Topic: Forces and Motion

Original examination-style practice material

Name: ______________________________	Date: __________________
Time allowed: 20 minutes	Total marks: 11

Show all necessary working. Give units where appropriate. Answer every question.

1 A trolley is pulled forwards with a force of 18 N. Friction is 6 N backwards.

(a) Calculate the resultant force. [2]

(b) State its direction. [1]

2 A 900 kg car has a driving force of 2400 N and a resistive force of 600 N.

(a) Calculate the resultant force. [2]

(b) Calculate the acceleration of the car. [3]

3 A parachutist is falling at terminal velocity. Explain what this means in terms of the forces and motion. [3]

End of Question Paper — check your answers.

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1 12 N forwards

Resultant force = 18 - 6 = 12 N.

Marks: subtracts opposing force · 12 N · forwards [3]

2 (a) 1800 N forwards (b) 2.0 m/s²

Resultant force = 2400 - 600 = 1800 N.

a = F/m = 1800/900 = 2.0 m/s².

Marks: subtracts forces · resultant · selects F = ma · substitution · answer with unit [5]

Accept: 2 m/s²

3 Weight equals air resistance, so the resultant force and acceleration are zero; the parachutist continues at constant velocity.

Marks: forces balanced · zero acceleration · constant velocity [3]